

PROJECT BUTTON HR–V0 ELECTRICAL V3 VISUAL–CORRECTION CANDIDATE

P1.18 connectivity preserved; page layout corrected for bounded labels, readable notes, and reviewable title blocks.

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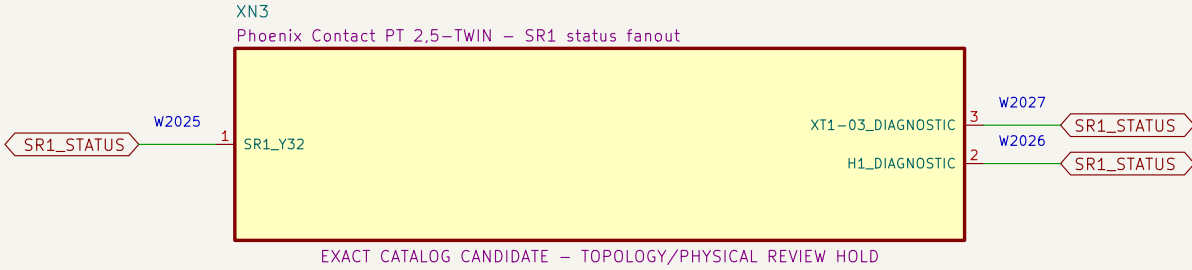
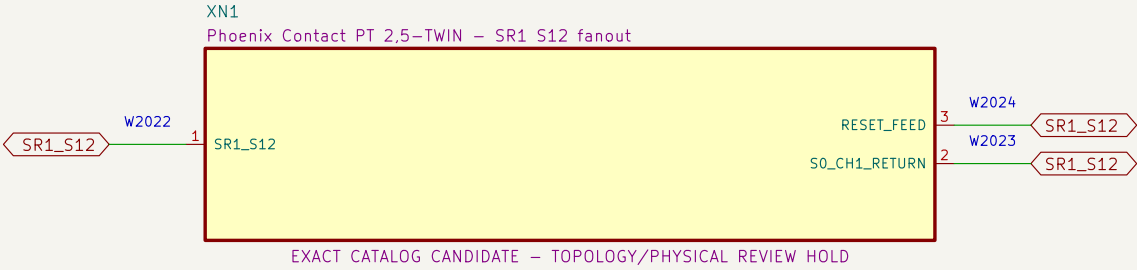
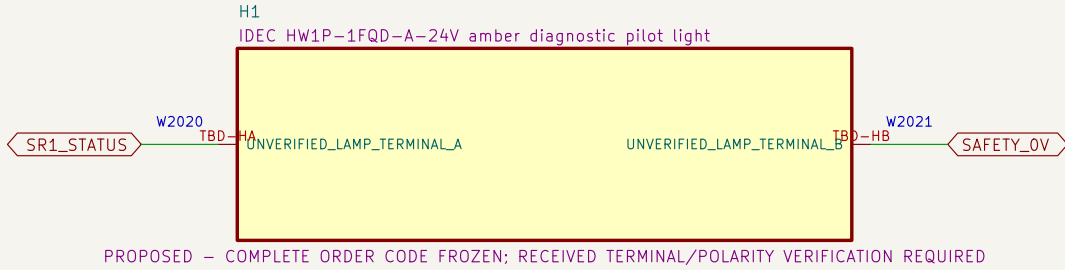
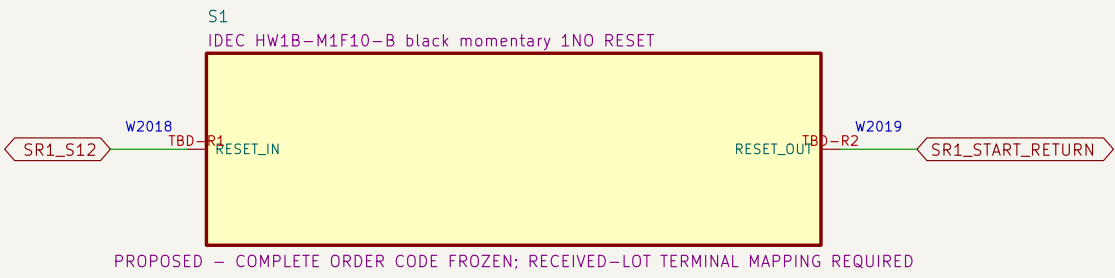
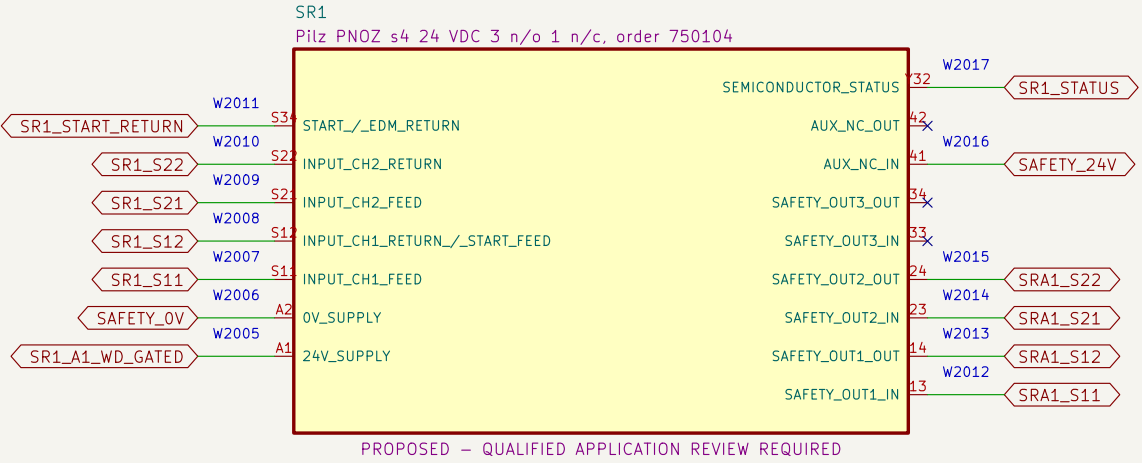
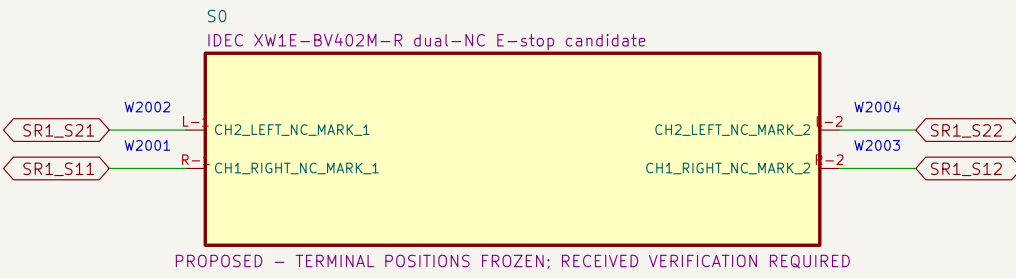
12 Watchdog PCB test access

File: 12_watchdog_pcb_test_access.kicad_sch

UNACCEPTED LAYOUT CANDIDATE SEE FULL PRELIMINARY WARNING ABOVE		
Project Button		
Sheet: / File: project–button–v3–p1.19–visual–correction–candidate.kicad_sch		
Title: PB HR–V0 ELEC P1.19 INDEX		
Size: A3	Date: 2026–08–11	Rev: P1.19
KiCad E.D.A. 10.0.5		Id: 1/13

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.

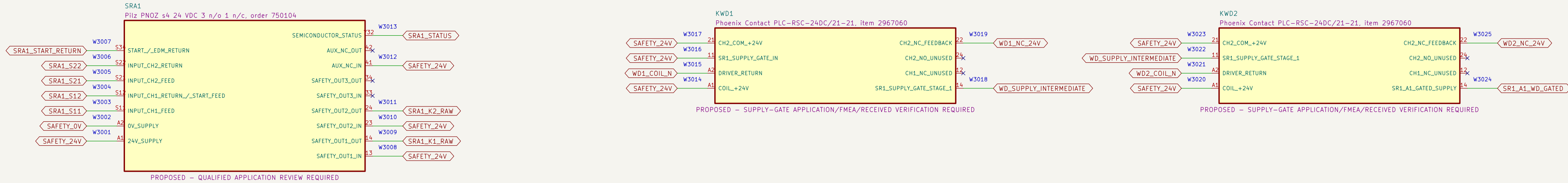


NOTE 1: S0 directly completes both SR1 input returns; ordinary KWD terminals are absent from both loops.

NOTE 2: Heartbeat loss removes SR1 A1 through the separate two-contact supply gate; recovery alone cannot restore the monitored RESET stage.

03 Distinct ARM and watchdog–gated SR1 supply

Two ordinary KWD contacts gate SR1 A1; SRA1 still requires SR1 outputs, EDM proof and a new ARM action.

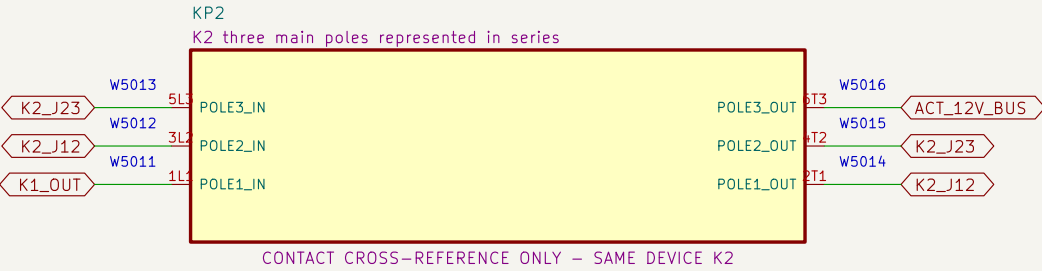
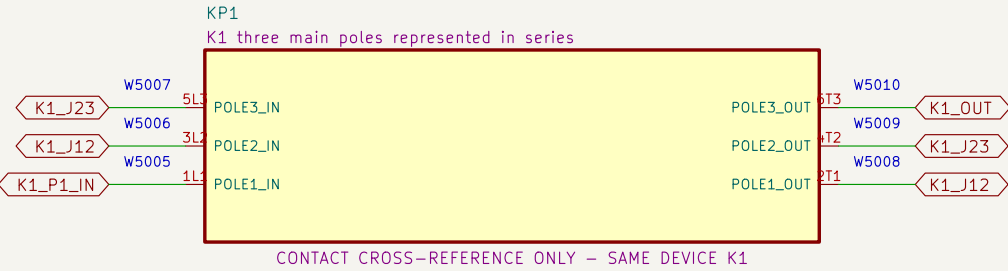
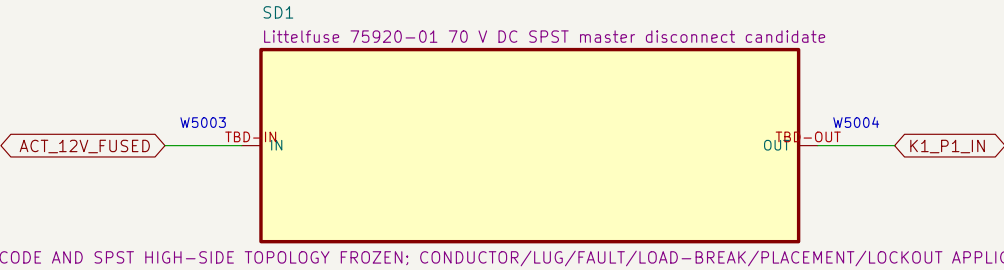
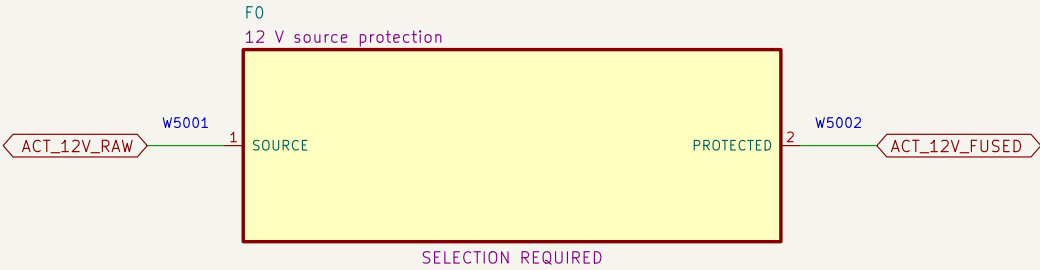


NOTE 1: Required after E–stop/watchdog dropout: cause healthy –> KWD supply restored –> RESET press/release –> SAFE_READY –> distinct ARM press/release.

NOTE 2: KWD contacts are in series with SR1 A1 only; S0 remains direct in both SR1 input returns and SRA1 receives both SR1 safety outputs directly.

05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



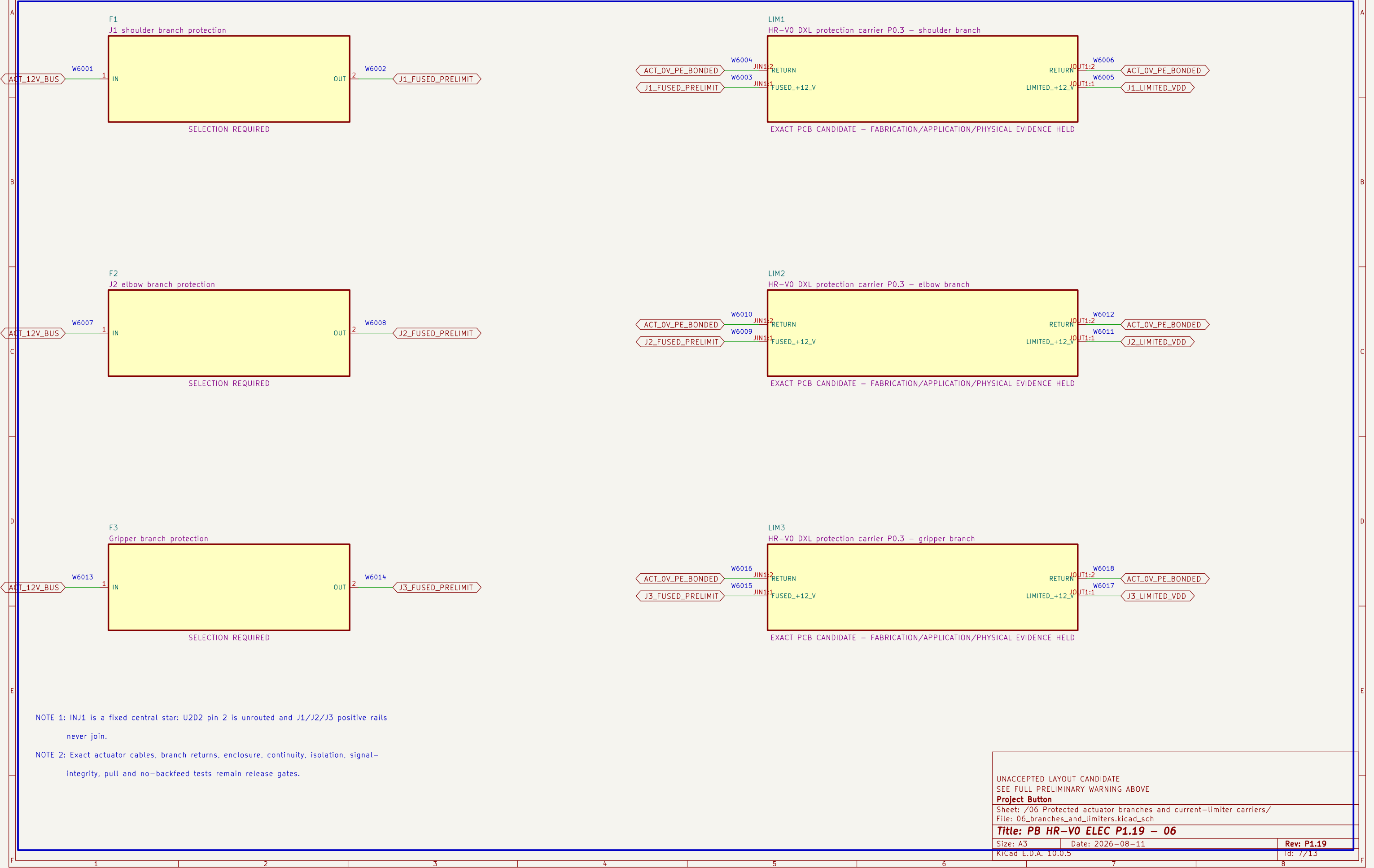
NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

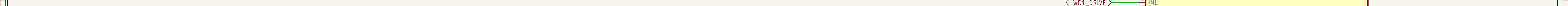
NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

UNACCEPTED LAYOUT CANDIDATE SEE FULL PRELIMINARY WARNING ABOVE		
Project Button		
Sheet: /05 Redundant actuator–power interruption/ File: 05_actuator_interruption.kicad_sch		
Title: PB HR–VO ELEC P1.19 – 05		
Size: A3	Date: 2026–08–11	Rev: P1.19
KiCad E.D.A. 10.0.5		Id: 6/13

06 Protected actuator branches and current-limiter carriers

Each fuse output and carrier output has a distinct positive rail; physical protection performance remains unproved.



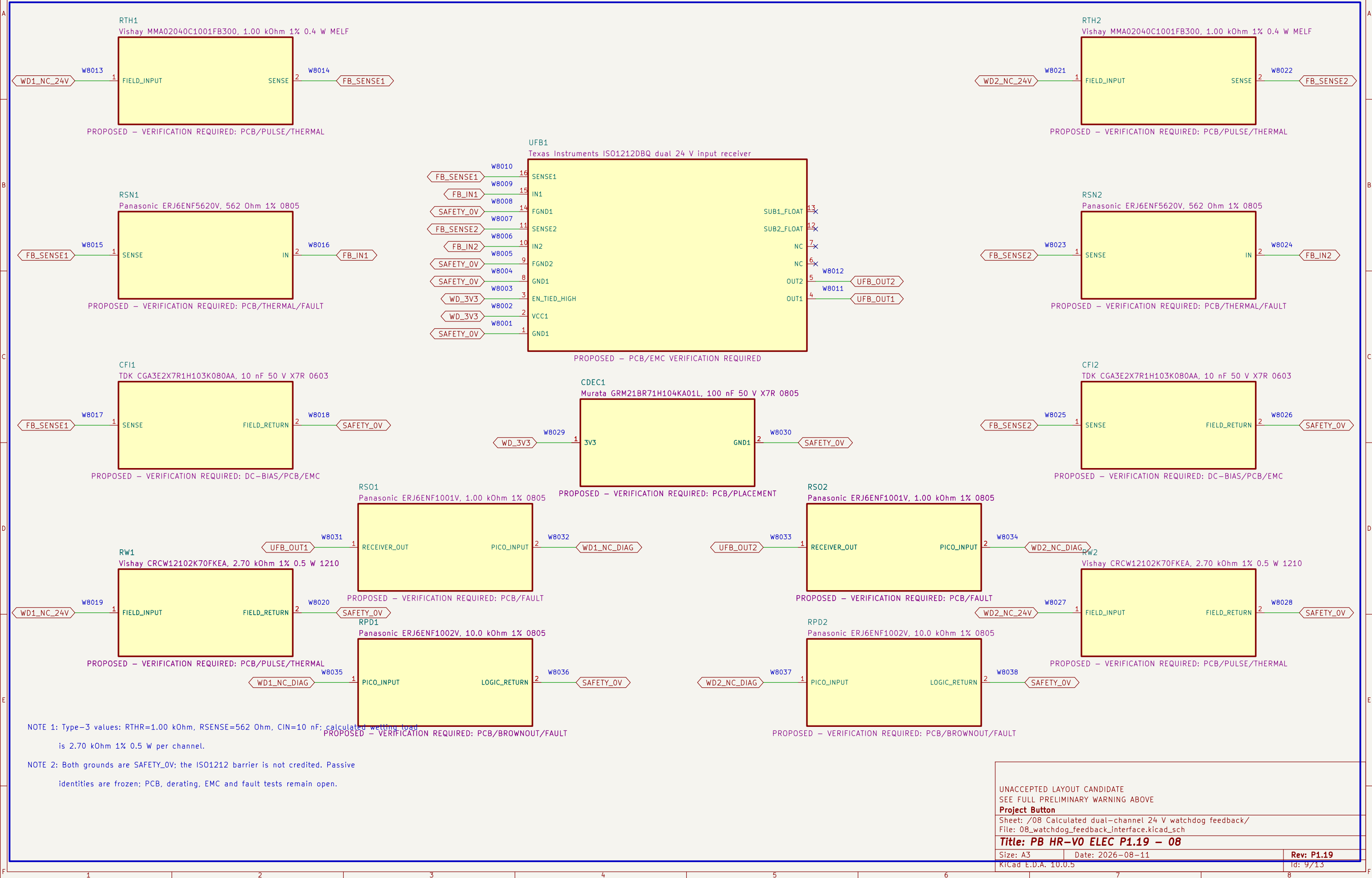


Title: RB HP-V0 ELEC P1 19 - 07

KIC80 L.D.A. 10.0.5 10. 0/15

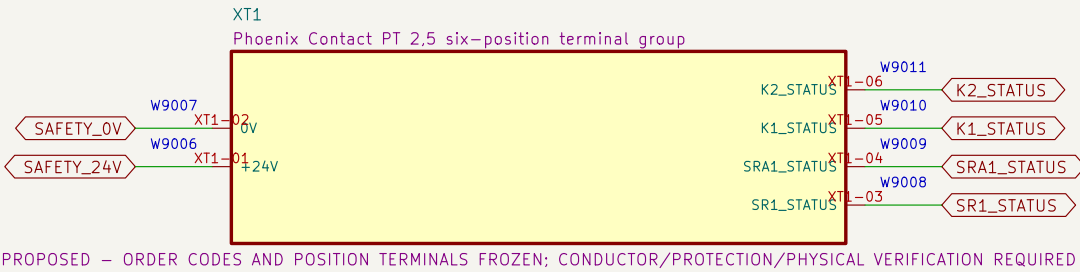
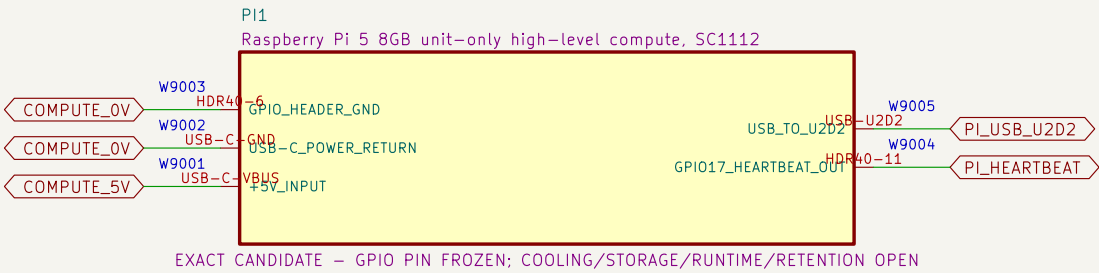
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.

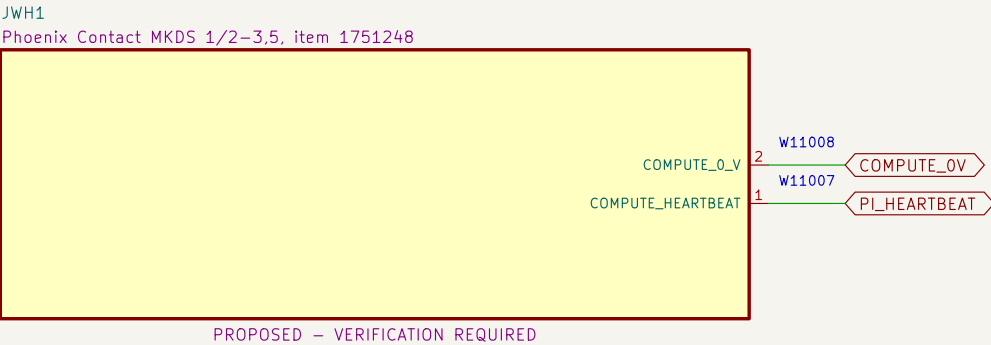
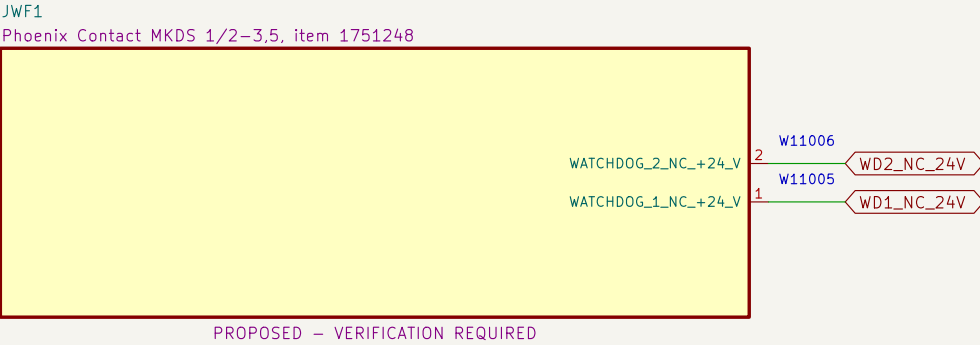
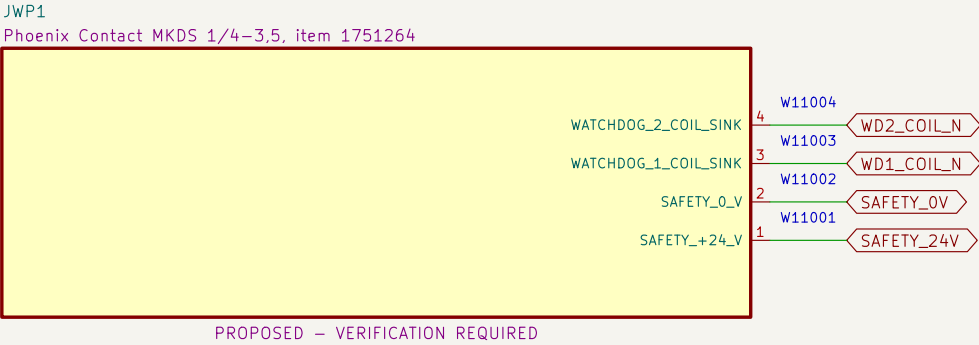


- NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2 under a future controlled unpowered-fixtured procedure.
- NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back-power the board or bypass a protective function.
- NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range, torque and enclosure layout remain selection required.

UNACCEPTED LAYOUT CANDIDATE SEE FULL PRELIMINARY WARNING ABOVE		
Project Button		
Sheet: /09 Compute and control terminals/ File: 09_compute_and_control_terminals.kicad_sch		
Title: PB HR-V0 ELEC P1.19 – 09		
Size: A3	Date: 2026-08-11	Rev: P1.19
KiCad E.D.A. 10.0.5		Id: 10/13

11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



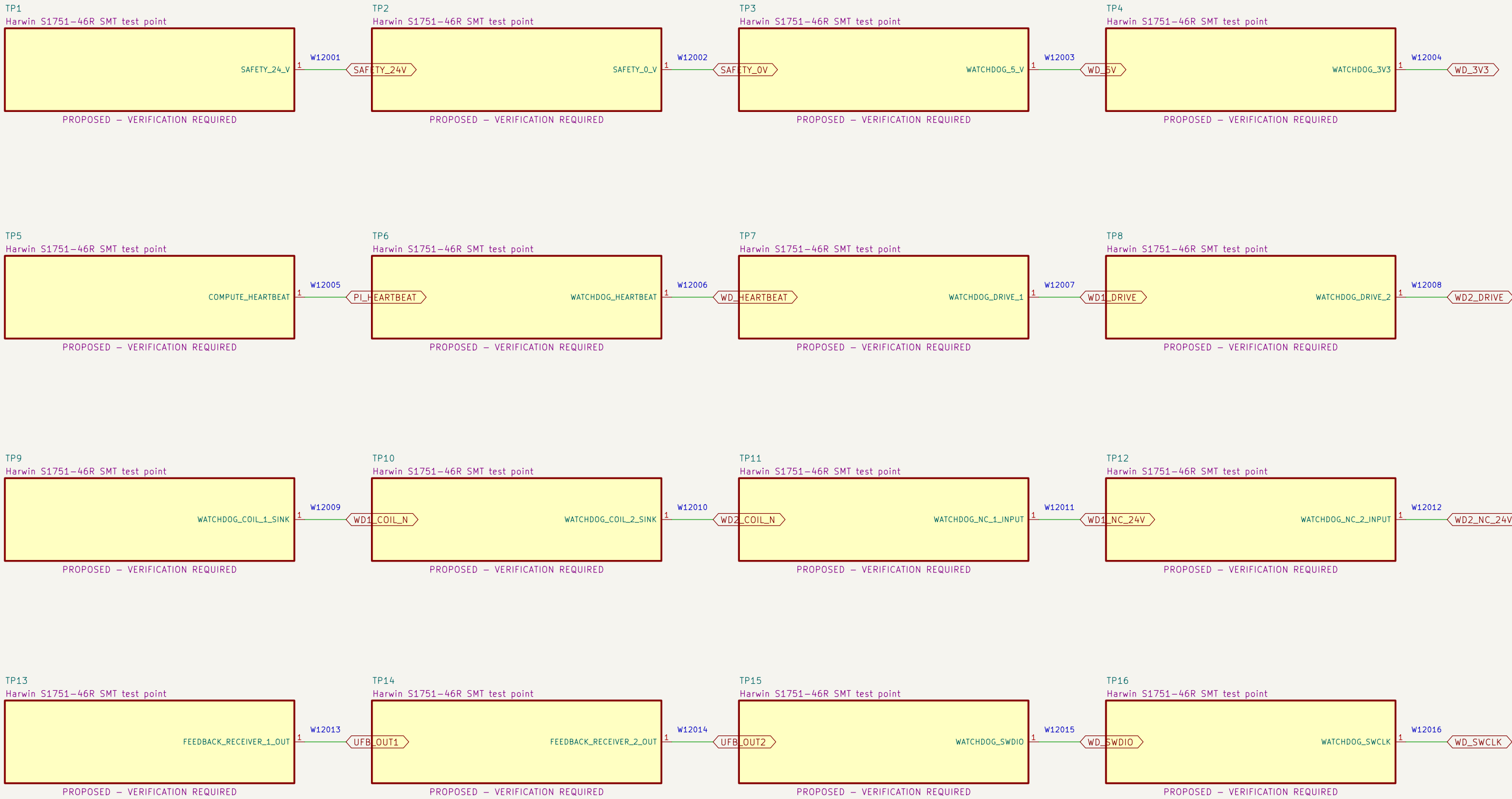
NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

UNACCEPTED LAYOUT CANDIDATE SEE FULL PRELIMINARY WARNING ABOVE		
Project Button		
Sheet: /11 Watchdog PCB external connectors/ File: 11_watchdog_pcb_connectors.kicad_sch		
Title: PB HR–VO ELEC P1.19 – 11		
Size: A3	Date: 2026–08–11	Rev: P1.19
KiCad E.D.A. 10.0.5	Id: 12/13	

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.